

From Convergence to Generalization: Stability of Stationary-Point Learning Algorithms

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Abstract—Algorithmic stability is a fundamental concept in learning theory for studying the generalization guarantees of learning algorithms. A notable limitation of classical stability analyses is that they often require convexity assumptions to obtain nontrivial bounds. In this paper, we investigate the stability and generalization properties of learning algorithms in nonconvex settings. We introduce an algorithm-dependent quantity that depends only on the training dataset and the algorithm's output. Under a mild differentiability assumption, we establish stability and generalization bounds that apply to almost any algorithm. Our bounds explicitly involve the optimization error and the algorithm-dependent quantity, thereby capturing the local curvature of the objective function around the learned model. A key feature of our analysis is that it remains valid even when the algorithm does not converge to a global or local minimizer. We further apply our general framework to gradient descent and demonstrate its implications for both linear models and shallow neural networks. Empirical studies verify the effectiveness of our stability analyses.

Index Terms—Learning theory, generalization bound, algorithmic stability, nonconvex learning.

I. INTRODUCTION

IN MACHINE learning (ML), models are typically trained by applying a learning algorithm to approximately maximize performance on a training dataset. However, good training performance does not necessarily translate into strong performance on unseen test data. In particular, when a model is overly complex, it may fit the training data well while failing to generalize to new samples. The discrepancy between training and testing performance is commonly referred to as the generalization gap. Understanding this generalization gap is a central topic in statistical learning theory (SLT) [1], [2]. A key challenge in this

direction is the dependence between the learned model and the training dataset [1], [2], due to which the training error evaluated at the output model is no longer an unbiased estimator of the testing error.

Various techniques have been developed to address this dependence. One of the most widely used approaches is the complexity-based analysis, which controls the uniform deviation between training and testing errors over a hypothesis space in which the learning process operates [3]. This approach typically yields generalization bounds that depend on measures of the hypothesis space complexity, such as the VC dimension, covering numbers, and Rademacher complexity [2], [3]. Another powerful framework for addressing this dependence is algorithmic stability, which quantifies the sensitivity of an algorithm to small perturbations of the training dataset. A foundational result shows that the generalization gap can be controlled by the algorithmic stability [4]. This insight has motivated substantial interest in stability-based analyses of learning algorithms, particularly in stochastic contexts [5]. Beyond these approaches, information-theoretic analyses capture the dependence via mutual information [6], [7], while PAC-Bayesian analyses bound the generalization gap for randomized predictors by controlling the KL divergence between posterior and prior distributions [8], [9]. Early developments in SLT primarily relied on complexity-based approaches, which often fail to explain the remarkable generalization performance of modern ML models. In practice, algorithms are frequently applied to highly overparameterized models, which, according to classical complexity measures, should severely overfit the training data. Contrary to this expectation, empirical evidence shows that such overparameterized models can achieve not only near-zero training error but also strong generalization performance [10]. These observations have spurred growing interest in alternative theoretical frameworks for understanding generalization, including algorithmic stability [5], PAC-Bayesian analysis [8], [9], and information-theoretic approaches [6], [7], [9]. Among these, algorithmic stability offers a particularly appealing perspective, as it explicitly incorporates the optimization trajectory into the generalization analysis. This often leads to tighter and complexity-independent generalization bounds [4], [5]. Such properties make stability-based analyses especially well-suited for overparameterized models, where the number of parameters exceeds the number of training samples.

Algorithmic stability has been widely used to study the generalization behavior of learning algorithms. However, most existing stability analyses rely on convexity assumptions to

Received 27 January 2026; revised 23 April 2026; accepted 24 April 2026. Date of publication 29 April 2026; date of current version 6 August 2026. The work of Yunwen Lei was supported in part by the Hong Kong Research Grants Council under the GRF projects under Grant 17302624 and Grant 17305425. The work of Zimeng Wang was supported in part by the Wallenberg AI, Autonomous Systems, and Software Program (WASP), funded by the Knut and Alice Wallenberg Foundation. The work of Xiaoming Yuan was supported in part by the Hong Kong Research Grants Council under the GRF project under Grant 17309824 and in part by Croucher Senior Fellowship. Recommended for acceptance by L. Shen. (*Corresponding author: Yunwen Lei.*)

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This article has supplementary downloadable material available at <https://doi.org/10.1109/TPAMI.2026.3688491>, provided by the authors.

Digital Object Identifier 10.1109/TPAMI.2026.3688491

obtain nontrivial bounds [5], [11], [12]. For example, the stability analysis of stochastic gradient descent (SGD) depends on the nonexpansiveness of the gradient operator [5], which holds only for convex objectives. This significantly limits the applicability of stability-based analysis. Recent work has sought to relax the convexity assumption. One line of research shows that, when the width m of a neural network is sufficiently large, the loss function exhibits a form of weak convexity, with the corresponding weak convexity parameter decaying on the order of $O(1/\sqrt{m})$ [13]. This observation has motivated stability analyses of gradient-based methods for training neural networks [14], [15]. Another line of work imposes the Polyak–Łojasiewicz (PL) condition and demonstrates that any learning algorithm converging to a global minimizer enjoys stability and generalization guarantees [16], [17]. While these relaxations of the convexity assumption significantly broaden the scope of stability analysis, several important limitations remain. For instance, the PL condition is still a strong requirement, as it implies that every stationary point is a global minimizer [18]. Moreover, existing analyses typically require the PL condition to hold uniformly over the entire parameter space [16], [17], which may not accurately reflect the behavior of the algorithm. In many cases, what truly matters is whether a PL-type condition holds along the optimization trajectory rather than globally. Even when the PL condition is satisfied globally, the associated PL constant may be too small to yield tight generalization bounds. Finally, current stability analyses under the PL condition [16], [17] guarantee nontrivial generalization mainly when the algorithm finds a global minimizer, a requirement that is often violated when solving nonconvex problems.

In this paper, we aim to advance the stability analysis of learning algorithms for nonconvex problems. Our main contributions are summarized as follows:

- We introduce an algorithm-dependent quantity as a relaxation of the PL parameter. Unlike the PL parameter, this quantity depends on the output of the algorithm and can be significantly larger, providing a more flexible measure for analyzing stability.
- We show that this algorithm-dependent quantity governs both the stability and generalization of learning algorithms. Our analysis explicitly captures the dependence of generalization on the optimization error and the curvature of the objective around the algorithm’s output. Notably, our results require only differentiability and symmetry¹ of the objective and apply to algorithms that output stationary points, in contrast to existing analyses [16], [17], which assume convergence to a global minimizer.
- We further refine the dependency on the optimization error by introducing a one-point quadratic-growth (QG) parameter, which extends and sharpens prior stability analyses under a QG condition [16] by removing several restrictive assumptions and yielding tighter bounds.

¹The symmetry assumption is not essential and is imposed here to simplify the presentation.

TABLE I
SUMMARY OF NOTATIONS

Notation	Meaning	Reference
β_S	PL parameter	Eq. (IV.1)
$\beta_{S,\mathbf{w}}$	$\frac{\ \nabla F_S(\mathbf{w})\ _2^2}{2 \max\{F_S(\mathbf{w}) - F_S(A(S)), 0\}}$	Eq. (IV.2)
$\beta'_{S,\mathbf{w}}$	$\frac{\ \nabla F_S(\mathbf{w})\ _2^2}{2(F_S(\mathbf{w}) - \tilde{F}_S)}$	Eq. (IV.3)
$\tilde{\beta}_{S,\mathbf{w}}$	a term $\leq \beta'_{S,\mathbf{w}}$	Theorem IV.11
$\tilde{\beta}_S$	$\tilde{\beta}_S \leq \tilde{\beta}_{S,\mathbf{w}_0}/2$	Theorem IV.11
μ_S	$\frac{2(F_S(A(S)) - \tilde{F}_S)}{\ A(S) - \tilde{A}(S)\ _2^2}$	Eq. (IV.10)
μ_{qg}	QG parameter	Eq. (IV.9)

- We apply our general results to gradient descent (GD) and derive excess risk bounds of order $O(1/n)$, where n is the sample size. We further demonstrate the practical utility of these bounds by improving existing generalization guarantees for GD applied to linear models and shallow neural networks.
- We conduct empirical analyses to validate our theoretical analyses. Experimental results show our stability bound is indeed an effective control of the generalization gap.

The remainder of the paper is organized as follows. We review related work in Section II and provide background in Section III. Our main results on stability analysis are presented in Section IV, followed by applications in Section V. Experimental results are reported in Section VI. Proofs of the main results are provided in Section VII. We conclude the paper in Section VIII, and defer all remaining experimental analyses and technical proofs to the supplementary material.

II. RELATED WORK

A. Stability and Generalization

Algorithmic stability quantifies how a small perturbation of the training dataset affects the output of a learning algorithm. One of the most widely studied notions is uniform stability [4], which can be used to derive high-probability generalization bounds [19], [20], [21]. Weaker stability notions have also been introduced to establish generalization guarantees under more relaxed conditions. For example, on-average stability studies generalization in expectation [22], and several other notions measure changes in the model parameters themselves, such as argument stability [23] and on-average model stability [12]. For randomized algorithms, stability can be quantified in terms of distances between the probability distributions of output models [24]. Typically, stability-based analyses yield generalization bounds of order $O(1/\sqrt{n})$. Subsequent work has established faster rates of $O(1/n)$ under additional assumptions, such as a low-noise condition [12], [25].

B. Stability Analysis of Gradient-Based Methods

An appealing feature of algorithmic stability is that it captures intrinsic properties of the learning algorithm and yields algorithm-dependent generalization bounds. The influential work [5] pioneered the stability analysis of SGD, which has since motivated numerous studies on the stability of other optimization algorithms [12], [16], [26], [27], [28], [29]. Optimistic stability bounds have been developed for gradient-based methods applied to both convex models [12] and neural networks [14], [15], [30], [31]. The stability of SGD in nonsmooth settings has also been analyzed [12], [23], while hypothesis stability [16] and on-average stability bounds [17] have been established for black-box optimization methods under the PL condition, which requires the convergence of an algorithm to a global minimizer. Algorithmic stability has been widely employed to study generalization in diverse settings, including meta-learning [32], [33], triplet learning [34], zeroth-order SGD [35], differential privacy [23], delayed SGD [36], biased gradient methods [37], AUPRC maximization [38] and federated learning [39].

III. BACKGROUND

Let ρ be a probability measure defined on a sample space $\mathcal{Z} := \mathcal{X} \times \mathcal{Y}$, where $\mathcal{X}$ is an input space and $\mathcal{Y} \subset \mathbb{R}$ is an output space. Suppose we draw a training dataset $S = \{\mathbf{z}_1, \dots, \mathbf{z}_n\}$ independently from ρ , and we want to build a model $h : \mathcal{X} \mapsto \mathcal{Y}$ for prediction. For brevity, we consider parametric models of the form $h_{\mathbf{w}}$, where $\mathbf{w}$ comes from a closed and convex parameter space $\mathcal{W} \subset \mathbb{R}^d$ (d is the dimensionality of the model). We measure the performance of a model $\mathbf{w}$ on a single example $\mathbf{z}$ via a loss function $f(\mathbf{w}; \mathbf{z})$. The performance on the training dataset and on the underlying distribution is quantified by the empirical risk $F_S : \mathcal{W} \rightarrow \mathbb{R}$ and the population risk $F : \mathcal{W} \rightarrow \mathbb{R}$, respectively:

$$F_S(\mathbf{w}) = \frac{1}{n} \sum_{i=1}^n f(\mathbf{w}; \mathbf{z}_i), \quad F(\mathbf{w}) = \mathbb{E}_{\mathbf{z}}[f(\mathbf{w}; \mathbf{z})],$$

where $\mathbb{E}_{\mathbf{z}}[\cdot]$ denotes the expectation with respect to $(\text{w.r.t.}) \mathbf{z}$. Throughout the paper, we assume that $f(\cdot; \mathbf{z})$ is differentiable for every $\mathbf{z} \in \mathcal{Z}$.

We often apply a learning algorithm $A : \cup_{n \in \mathbb{N}} \mathcal{Z}^n \mapsto \mathcal{W}$ to train a model $\mathbf{w} \in \mathcal{W}$ based on a training dataset S . We denote by $A(S)$ the model produced by applying A to S , which may be stochastic (e.g., SGD) or deterministic (e.g., GD). After obtaining $A(S)$, we are interested in its performance relative to the optimal model $\mathbf{w}^* = \arg \min_{\mathbf{w} \in \mathcal{W}} F(\mathbf{w})$. A standard approach is to decompose the excess population risk [40] as

$$\begin{aligned} \mathbb{E}[F(A(S))] - F(\mathbf{w}^*) &= \mathbb{E}[F(A(S)) - F_S(A(S))] \\ &\quad + \mathbb{E}[F_S(A(S)) - F_S(\mathbf{w}^*)], \end{aligned}$$

where we have used the equality $\mathbb{E}[F_S(\mathbf{w}^*)] = F(\mathbf{w}^*)$ since $\mathbf{w}^*$ is independent of S . We refer to the first term as the *generalization gap*, which measures the difference between the training and testing performance of $A(S)$, and the second term as the *optimization error*, which quantifies the suboptimality of $A(S)$ in terms of the training loss. In this work, we leverage algorithmic stability to control the generalization gap, while

the optimization error is analyzed using standard tools from optimization theory.

We say that two datasets S and $\tilde{S}$ are neighboring datasets if they differ in at most one example. Below, we recall the definition of uniform stability [4].

Definition III.1 (Uniform stability): We say that an algorithm A is ϵ_{unif} -uniformly stable ($\epsilon_{\text{unif}} > 0$) if, for any two neighboring datasets S and $\tilde{S}$ we have $\sup_{\mathbf{z}} \mathbb{E}_A[|f(A(S); \mathbf{z}) - f(A(\tilde{S}); \mathbf{z})|] \leq \epsilon_{\text{unif}}$, where $\mathbb{E}_A[\cdot]$ denotes the expectation taken w.r.t. to the randomness of A .

Uniform stability can be somewhat restrictive since it involves a supremum over both $\mathbf{z}$ and all neighboring datasets. In this paper, we instead consider on-average stability, which measures the average effect of perturbing each training example in the dataset. Let $S = \{\mathbf{z}_1, \dots, \mathbf{z}_n\}$ and $S' = \{\mathbf{z}'_1, \dots, \mathbf{z}'_n\}$ be two datasets of size n drawn independently from ρ . For each $i \in [n] := \{1, \dots, n\}$, define

$$S_i = \{\mathbf{z}_1, \dots, \mathbf{z}_{i-1}, \mathbf{z}'_i, \mathbf{z}_{i+1}, \dots, \mathbf{z}_n\}. \quad (\text{III.1})$$

Clearly, S_i is a neighboring dataset of S .

Definition III.2 (On-average stability): Let S, S' be two datasets of size n drawn independently from ρ , and let S_i be defined as in (III.1) for each $i \in [n]$. We say that an algorithm A is on-average ϵ_A -stable ($\epsilon_A > 0$) if

$$\frac{1}{n} \sum_{i=1}^n \mathbb{E}_{S, S'} [f(A(S_i); \mathbf{z}_i) - f(A(S); \mathbf{z}_i)] \leq \epsilon_A.$$

The following lemma indicates that on-average stability guarantees generalization in expectation. Note that ϵ_A becomes a random variable if A is a randomized algorithm.

Lemma III.3 ([22]): If an algorithm A is on-average ϵ_A -stable, then $\mathbb{E}_S[F(A(S)) - F_S(A(S))] \leq \epsilon_A$.

We also recall the definitions of Lipschitz continuity, smoothness, and strong convexity as follows.

Definition III.4 (Smoothness, Lipschitzness, and strong convexity): Let $\mu, L, G \geq 0$. For a differentiable function $g : \mathcal{W} \mapsto \mathbb{R}$, we say g is L -smooth if

$$\|\nabla g(\mathbf{w}) - \nabla g(\mathbf{w}')\|_2 \leq L \|\mathbf{w} - \mathbf{w}'\|_2, \quad \forall \mathbf{w}, \mathbf{w}' \in \mathcal{W}.$$

We say g is G -Lipschitz continuous if $\|\nabla g(\mathbf{w})\|_2 \leq G$ for any $\mathbf{w} \in \mathcal{W}$. Moreover, we say g is μ -strongly convex if for any $\mathbf{w}, \mathbf{w}' \in \mathcal{W}$,

$$g(\mathbf{w}) \geq g(\mathbf{w}') + \langle \mathbf{w} - \mathbf{w}', \nabla g(\mathbf{w}') \rangle + \frac{\mu}{2} \|\mathbf{w} - \mathbf{w}'\|_2^2.$$

We say g is convex if the above inequality holds with $\mu = 0$.

IV. MAIN RESULTS

A. Algorithm-Dependent Generalization Bounds

In this subsection, we present a general result characterizing the generalization behavior of learning algorithms that converge to stationary points. First, recall the Polyak-Łojasiewicz (PL) condition as follows. We collect notations on the landscape of problems in Table 1.

Definition IV.1 (PL condition): We say that a differentiable function $F_S : \mathcal{W} \rightarrow \mathbb{R}$ satisfies the PL condition with parameter

$\beta_S > 0$ if

$$F_S(\mathbf{w}) - \hat{F}_S \leq \frac{1}{2\beta_S} \|\nabla F_S(\mathbf{w})\|_2^2, \quad \forall \mathbf{w} \in \mathcal{W}, \quad (\text{IV.1})$$

where $\hat{F}_S := \min_{\mathbf{w} \in \mathcal{W}} F_S(\mathbf{w})$.

Remark IV.2: The PL condition is widely used to study the convergence of gradient-based methods [18]. The work [16] identified its role in analyzing the stability of learning algorithms, showing that the hypothesis stability of any algorithm can be controlled by the PL parameter β_S together with the optimization error. This result was further refined in [17], which leveraged the smoothness of the loss functions to derive optimistic on-average stability bounds under the PL condition.

Note that the PL condition is a global property and is thus relatively strong. To increase the applicability of our results, we introduce two weaker notations next. For any S and $\mathbf{w} \in \mathcal{W}$, define $\beta_{S,\mathbf{w}} \in (0, +\infty]$ as

$$\beta_{S,\mathbf{w}} := \frac{\|\nabla F_S(\mathbf{w})\|_2^2}{2 \max\{F_S(\mathbf{w}) - F_S(A(S)), 0\}}, \quad (\text{IV.2})$$

where $\|\cdot\|_2$ denotes the Euclidean norm and $\nabla F_S : \mathcal{W} \rightarrow \mathbb{R}^d$ is the gradient of F_S . Here we use the maximum operator to ensure that $\beta_{S,\mathbf{w}}$ is nonnegative. Additionally, we define

$$\beta'_{S,\mathbf{w}} := \frac{\|\nabla F_S(\mathbf{w})\|_2^2}{2(F_S(\mathbf{w}) - \hat{F}_S)}. \quad (\text{IV.3})$$

Note $\beta_{S,\mathbf{w}} \geq \beta'_{S,\mathbf{w}}$ for all S and $\mathbf{w}$ due to $\hat{F}_S \leq F_S(A(S))$. Besides, $\beta'_{S,\mathbf{w}}$ is closely related to the PL condition. For a fixed S , if F_S satisfies the PL condition with $\beta_S > 0$, then $\beta'_{S,\mathbf{w}} \geq \beta_S$ for all $\mathbf{w} \in \mathcal{W}$. A notable property is that $\beta'_{S,\mathbf{w}}$ depends on both S and $\mathbf{w}$, the latter of which is often chosen as the output of the algorithm applied to S in the generalization analysis (e.g., we choose $\mathbf{w} = A(S)$). Therefore, $\beta'_{S,\mathbf{w}}$ can be much larger than the PL parameter β_S , and hence so is $\beta_{S,\mathbf{w}}$. In the supplementary material, we provide empirical evidence demonstrating that $\beta_{S,\mathbf{w}}$ is significantly larger than β_S when training shallow neural networks (SNNs).

Next, we present our main result in the following theorem. Our analysis clarifies the crucial role of the quantity $\beta_{S,\mathbf{w}}$ on the generalization, which relaxes the PL condition. Some discussions also consider the local PL condition, where (IV.1) holds around a neighborhood of the initial point $\mathbf{w}_0$ [13]. Geometrically, this implies that any point with a small gradient must lie close (in function value) to the global minimum. In contrast, $\beta_{S,\mathbf{w}}$ compares the gradient to the gap relative to the algorithmic output, i.e., $F_S(\mathbf{w}) - F_S(A(S))$. This leads to two key distinctions. First, if $\beta_{S,\mathbf{w}}$ is bounded from below, points with small gradients can still be far from the global minimum, provided they remain close to the level set of $F_S(A(S))$. Second, while the local PL condition requires (IV.1) to hold throughout a local neighborhood, our stability bound in Theorem IV.3 only requires $\beta_{S_1,A(S)}$ at the specific point $A(S)$. This makes our approach more flexible for analyzing algorithms that converge to stationary points rather than global minimizers. In the remainder of the paper, when we write $\mathbb{E}[\cdot]$, we mean the expectation taken w.r.t. S, S' and A . We always assume A is symmetric, i.e., the distribution of $A(S)$ does not depend on the order of examples

in S . This is a mild assumption that holds for a wide range of algorithms, including GD and SGD.

Theorem IV.3: Let S_i be defined in (III.1) and A be a symmetric algorithm. Let $\beta_{S,\mathbf{w}}$ be defined in (IV.2). Then A is on-average ϵ_A -stable with

$$\epsilon_A = \mathbb{E} \left[\frac{2\|\nabla f(A(S); \mathbf{z}_1)\|_2^2 + 2\|\nabla f(A(S); \mathbf{z}'_1)\|_2^2}{n\beta_{S_1,A(S)}} \right] + \mathbb{E} \left[\frac{n\|\nabla F_S(A(S))\|_2^2}{\beta_{S_1,A(S)}} \right]. \quad (\text{IV.4})$$

If F_S is L_S -smooth, then A is on-average ϵ_A -stable with

$$\epsilon_A = \mathbb{E} \left[\frac{2\|\nabla f(A(S); \mathbf{z}_1)\|_2^2 + 2\|\nabla f(A(S); \mathbf{z}'_1)\|_2^2}{n\beta_{S_1,A(S)}} \right] + \mathbb{E} \left[\frac{2L_S n(F_S(A(S)) - \hat{F}_S)}{\beta_{S_1,A(S)}} \right]. \quad (\text{IV.5})$$

Furthermore, we have $\mathbb{E}[F(A(S)) - F_S(A(S))] \leq \epsilon_A$.

Remark IV.4 (Applicability to algorithms finding stationary points): Theorem IV.3 gives stability and generalization bounds for general learning algorithms in terms of $\beta_{S_1,A(S)}$. Eq. (IV.4) is derived without any assumptions except the differentiability of the loss function. It shows that the stability can be bounded by two terms: $\mathbb{E}[(\|\nabla f(A(S); \mathbf{z}_1)\|_2^2 + \|\nabla f(A(S); \mathbf{z}'_1)\|_2^2)/(n\beta_{S_1,A(S)})]$ and $n\mathbb{E}[\|\nabla F_S(A(S))\|_2^2/\beta_{S_1,A(S)}]$. The first term is controlled by the sample size n and the landscape of the problem (reflected by $\beta_{S_1,A(S)}$). On the other hand, the second term is related to the optimization error, which would typically be smaller if we run more iterations. A key difference between our results and the existing stability analysis [16], [17] is that our analysis does not require the algorithm to find a global minimizer. The reason is that (IV.4) involves the optimization error $\|\nabla F_S(A(S))\|_2^2$, while the generalization bounds in [16], [17] involve $F_S(A(S)) - \hat{F}_S$. Generally, optimization algorithms only guarantee a stationary point for nonconvex problems since the optimization error is measured by the magnitude of the gradients [41]. In this case, (IV.4) still applies, while the existing stability analysis may not. In particular, if A returns a stationary point, then (IV.4) implies that

$$\epsilon_A = \mathbb{E} \left[\frac{2\|\nabla f(A(S); \mathbf{z}_1)\|_2^2 + 2\|\nabla f(A(S); \mathbf{z}'_1)\|_2^2}{n\beta_{S_1,A(S)}} \right]. \quad (\text{IV.6})$$

For GD, we can show that $\|\nabla F_S(A(S))\|_2^2$ decays exponentially fast w.r.t. the number of iterations. Then, we can choose a moderate iteration number to make the second term in (IV.4) dominated by the first term. In this case, we get a generalization bound of the order of $\mathbb{E}[\|\nabla f(A(S); \mathbf{z}'_1)\|_2^2/(n\beta_{S_1,A(S)})]$. In the supplementary material, we empirically verify that $\frac{n\|\nabla F_S(A(S))\|_2^2}{\beta_{S_1,A(S)}}$ remains small when training SNNs with GD.

Remark IV.5 (Comparison with PL condition): Eq. (IV.5) is an extension of the generalization bound developed in [17]. If f is L -smooth with $L \leq n\beta/4$ and

$$\mathbb{E}[F_S(\mathbf{w}) - \hat{F}_S] \leq \frac{1}{2\beta} \mathbb{E}[\|\nabla F_S(\mathbf{w})\|_2^2], \quad \forall \mathbf{w} \in \mathcal{W}, \quad (\text{IV.7})$$

the following bound was established in [17]²

$$\mathbb{E}[F(A(S)) - \hat{F}_S] \lesssim \frac{L\mathbb{E}[\hat{F}_S]}{n\beta} + \frac{L\mathbb{E}[F_S(A(S)) - \hat{F}_S]}{\beta}. \quad (\text{IV.8})$$

Note that (IV.7) requires a PL parameter β that holds uniformly for all datasets S and all $\mathbf{w} \in \mathcal{W}$. In contrast, (IV.2) allows the parameter $\beta_{S,A(S)}$ to depend on both the dataset S and the algorithm's output $A(S)$, making it data-dependent. Note $\beta_{S,A(S)}$ is always lower bounded by β and could be significantly larger than β since we have $\beta \leq \min_{\mathbf{w} \in \mathcal{W}} \frac{\mathbb{E}[\|\nabla F_S(\mathbf{w})\|_2^2]}{2(\mathbb{E}[F_S(\mathbf{w}) - \hat{F}_S])}$. We shall give explicit estimates of $\beta_{S,A(S)}$ for linear models and SNNs in Section V. Furthermore, the analysis in [17] requires an assumption $L \leq n\beta/4$, which is removed in our analysis. It should be noted that (IV.8) features a more favorable dependence on the optimization error, as it scales with $L\mathbb{E}[F_S(A(S)) - \hat{F}_S]/\beta$, whereas (IV.5) scales with $nL\mathbb{E}[(F_S(A(S)) - \hat{F}_S)/\beta_{S_1,A(S)}]$, introducing an extra factor of n . However, this factor does not significantly affect algorithms operating in an interpolation setting. For instance, both SGD and GD are known to achieve linear convergence rates under interpolation [42]. Then, one can simply increase the number of iterations by a logarithmic factor to compensate for the additional n factor.

Remark IV.6: Note that the bound in (IV.4) is algorithm-dependent since it involves $A(S)$. Furthermore, because $\mathbb{E}[F_S(A(S))] = \mathbb{E}[F_{S_1}(A(S_1))]$, the term $1/\beta_{S_1,A(S)}$ can be approximated as

$$\begin{aligned} \frac{1}{\beta_{S_1,A(S)}} &= \frac{2 \max\{F_{S_1}(A(S)) - F_{S_1}(A(S_1)), 0\}}{\|\nabla F_{S_1}(A(S))\|_2^2} \\ &\approx \frac{2 \max\{F_{S_1}(A(S)) - F_S(A(S)), 0\}}{\|\nabla F_{S_1}(A(S))\|_2^2}. \end{aligned}$$

Since $A(S)$ has already been computed by the algorithm, both $F_S(A(S))$ and $F_{S_1}(A(S))$ can be evaluated directly.

Next, we consider the quadratic growth (QG) condition, based on which we introduce a relaxed quantity that leads to improved bounds.

Definition IV.7 (QG condition): We say F_S satisfies the quadratic growth condition with parameter $\mu_{qg} \geq 0$ if

$$F_S(\mathbf{w}) - \hat{F}_S \geq \frac{\mu_{qg}}{2} \|\mathbf{w} - \pi_S(\mathbf{w})\|_2^2, \quad \mathbf{w} \in \mathcal{W}, \quad (\text{IV.9})$$

where $\pi_S : \mathcal{W} \rightarrow \mathcal{W}$ denotes the Euclidean projection onto the set of global minimizers of F_S , i.e.,

$$\pi_S(\mathbf{w}) := \arg \min_{\tilde{\mathbf{w}} \in \mathcal{W}} \|\mathbf{w} - \tilde{\mathbf{w}}\|_2, \quad \text{s.t. } F_S(\tilde{\mathbf{w}}) = \hat{F}_S.$$

The QG condition is weaker than the PL condition. Indeed, it has been shown that if a function satisfies the PL condition, then the QG condition holds with the same parameter [18]. Similar to the PL condition, the QG condition is also a global property. To relax it, we introduce the one-point QG parameter $\mu_S \geq 0$ defined by

$$\mu_S := \frac{2(F_S(A(S)) - \hat{F}_S)}{\|A(S) - \tilde{A}(S)\|_2^2}, \quad (\text{IV.10})$$

²We denote $A \lesssim B$ if there exists a universal constant $c > 0$ such that $A \leq cB$. We denote $A \gtrsim \text{rsim} B$ if there exists a universal constant $c' > 0$ such that $A \geq c'B$. We denote $A \asymp B$ if $A \lesssim B$ and $A \gtrsim \text{rsim} B$.

where the algorithm $\tilde{A}$ projects the output of A onto the set of global minimizers, i.e.,

$$\tilde{A}(S) := \pi_S(A(S)). \quad (\text{IV.11})$$

Note that $\tilde{A}$ is symmetric if A is symmetric. Eq. (IV.10) gives a localization of (IV.9) at $\mathbf{w} = A(S)$ and hence $\mu_S \geq \mu_{qg}$. In the supplementary material, we provide empirical verification to show that μ_S can be significantly larger than μ_{qg} .

In Theorem IV.8, we leverage μ_S to derive a population risk bound with improved dependency on the optimization error.

Theorem IV.8: Let S_i be defined in (III.1) and A be a symmetric algorithm. Let $\beta_{S,\mathbf{w}}$ be defined in (IV.2). If the map $\mathbf{w} \mapsto f(\mathbf{w}; \mathbf{z})$ is L -smooth for any $\mathbf{z}$, then

$$\begin{aligned} \mathbb{E}[F(A(S)) - \hat{F}_S] &\lesssim \mathbb{E} \left[\frac{\|\nabla f(A(S); \mathbf{z}_1)\|_2^2 + \|\nabla f(A(S); \mathbf{z}'_1)\|_2^2}{n\beta_{S_1, \tilde{A}(S)}} \right] \\ &\quad + L\mathbb{E} \left[\|A(S) - \tilde{A}(S)\|_2^2 \left(1 + \frac{L}{n\beta_{S_1, \tilde{A}(S)}} \right) \right]. \quad (\text{IV.12}) \end{aligned}$$

Moreover, consider μ_S defined in (IV.10), then

$$\begin{aligned} \mathbb{E}[F(A(S)) - \hat{F}_S] &\lesssim \mathbb{E} \left[\frac{\|\nabla f(A(S); \mathbf{z}_1)\|_2^2 + \|\nabla f(A(S); \mathbf{z}'_1)\|_2^2}{n\beta_{S_1, \tilde{A}(S)}} \right] \\ &\quad + L\mathbb{E} \left[\left(\frac{F_S(A(S)) - \hat{F}_S}{\mu_S} \right) \left(1 + \frac{L}{n\beta_{S_1, \tilde{A}(S)}} \right) \right]. \quad (\text{IV.13}) \end{aligned}$$

Remark IV.9 (Comparison with QG condition): Theorem IV.8 improves upon Theorem IV.3 by replacing $n(F_S(A(S)) - \hat{F}_S)$ with $F_S(A(S)) - \hat{F}_S$. Therefore, it requires less accuracy in optimization to achieve similar risk bounds. If we further assume $L \lesssim n\beta_{S_1, \tilde{A}(S)}$, then (IV.13) implies

$$\begin{aligned} \mathbb{E}[F(A(S)) - \hat{F}_S] &\lesssim L\mathbb{E} \left[\frac{F_S(A(S)) - \hat{F}_S}{\mu_S} \right] \\ &\quad + \mathbb{E} \left[\frac{\|\nabla f(A(S); \mathbf{z}_1)\|_2^2 + \|\nabla f(A(S); \mathbf{z}'_1)\|_2^2}{n\beta_{S_1, \tilde{A}(S)}} \right]. \quad (\text{IV.14}) \end{aligned}$$

Under a QG condition and a G -Lipschitzness condition, together with the assumptions that $f(\mathbf{w}; \mathbf{z}) \leq c$ for all $\mathbf{w}, \mathbf{z}$ and $\pi_S(A(S)) = \pi_S(A(\tilde{S}))$ for neighboring datasets $S, \tilde{S}$, the following uniform stability bound was developed in [16]

$$\epsilon_{\text{unif}} \lesssim G \left(\frac{F_S(A(S)) - \hat{F}_S}{\mu_{qg}} \right)^{\frac{1}{2}} + G \left(\frac{c}{\mu_{qg}n} \right)^{\frac{1}{2}}. \quad (\text{IV.15})$$

We compare this bound with (IV.14) from three aspects. First, (IV.15) only guarantees a bound of order $1/\sqrt{n}$, while (IV.14) shows a faster bound of order $O(1/n)$. Second, (IV.15) requires the QG parameter μ_{qg} to hold for any $\mathbf{w} \in \mathcal{W}$. As a comparison, (IV.14) only involves μ_S depending on $A(S)$,

which could be much larger than μ_{qg} in practice. Third, (IV.15) requires additional assumptions such as the boundedness of f and $\pi_S(A(S)) = \pi_S(A(\tilde{S}))$ for any neighboring $S, \tilde{S}$. In particular, the latter assumption almost means that there is a unique global minimizer, which is restrictive in nonconvex settings. We remove these assumptions entirely, and obtain bounds involving potentially large parameters $\beta_{S_1, \tilde{A}(S)}$ and μ_S .

B. Generalization Bounds of Gradient Descent

We proceed to apply the general results obtained in Section IV-A to study the generalization performance of models trained by gradient descent (GD). We assume $\mathcal{W} = \mathbb{R}^d$ hereafter.

Definition IV.10 (Gradient Descent): Let $\mathbf{w}_0 \in \mathbb{R}^d$ and $\eta > 0$ be the step size. Given a dataset S , GD updates the model via

$$\mathbf{w}_{t+1} = \mathbf{w}_t - \eta \nabla F_S(\mathbf{w}_t), \quad t \geq 0. \quad (\text{GD})$$

To conduct analysis, we introduce a new quantity $\tilde{\beta}_{S, \mathbf{w}}$, which is no larger than $\beta'_{S, \mathbf{w}}$ given by (IV.3) for all S and $\mathbf{w} \in \mathbb{R}^d$. We impose an assumption that the parameter $\tilde{\beta}_{S, \mathbf{w}}$ is Lipschitz continuous w.r.t. $\mathbf{w}$ and a constraint on $\tilde{\beta}_{S, \mathbf{w}_0}$ that aims to make $\beta'_{S, \mathbf{w}}$ larger than $\tilde{\beta}_S$ in a neighborhood around $\mathbf{w}_0$ for some $\tilde{\beta}_S > 0$. In Section V, we shall verify these requirements with appropriately defined $\tilde{\beta}_{S, \mathbf{w}}$ for both linear models and SNNs. Theorem IV.11 gives population risk bounds for (GD) after a logarithmic number of iterations $T > 0$. Note that T benefits from large $\tilde{\beta}_S$. This parameter is related to $\tilde{\beta}_{S, \mathbf{w}_0}$, which captures the curvature at $\mathbf{w}_0$ and could be much larger than the PL parameter in Definition IV.1.

Theorem IV.11: Let $\mathbf{w}_0 \in \mathbb{R}^d$ be an initial model. For any $S, \mathbf{w}$, let $\beta'_{S, \mathbf{w}}$ be defined by (IV.3) and $\tilde{\beta}_{S, \mathbf{w}} \leq \beta'_{S, \mathbf{w}}$ for some $\tilde{\beta}_{S, \mathbf{w}} > 0$. Assume there exists $L_\beta > 0$ such that

$$|\tilde{\beta}_{S, \mathbf{w}} - \tilde{\beta}_{S, \mathbf{w}'}| \leq L_\beta \|\mathbf{w} - \mathbf{w}'\|_2. \quad (\text{IV.16})$$

Let $\tilde{\beta}_S > 0$ be chosen such that $\tilde{\beta}_S \leq \tilde{\beta}_{S, \mathbf{w}_0}/2$. Suppose $L_\beta \sqrt{8(F_S(\mathbf{w}_0) - \hat{F}_S)/\tilde{\beta}_S} \leq \tilde{\beta}_S$ and F_S is L_S -smooth. Let $\{\mathbf{w}_t\}_{t \in \mathbb{N}}$ be produced by (GD) with $\eta = 1/L_S$ and $A(S) = \mathbf{w}_T$. If $T \geq \frac{L_S}{\tilde{\beta}_S} \log(n^2 L_S (F_S(\mathbf{w}_0) - \hat{F}_S))$, then

$$\begin{aligned} & \mathbb{E}[F(A(S)) - \hat{F}_S] \\ & \lesssim \frac{1}{n^2 L_S} + \frac{1}{n} \mathbb{E} \left[\frac{\|\nabla f(A(S); \mathbf{z}_1)\|_2^2 + \|\nabla f(A(S); \mathbf{z}'_1)\|_2^2 + 1}{\beta_{S_1, A(S)}} \right]. \end{aligned} \quad (\text{IV.17})$$

The next theorem derives similar risk bounds for (GD) with fewer iterations by using the one-point QG parameter μ_S defined in (IV.10). Additionally, the risk bounds involve $\beta_{S_1, \tilde{A}(S)}$ rather than $\beta_{S_1, A(S)}$.

Theorem IV.12: Consider the same setting as in Theorem IV.11. Let μ_S be defined in (IV.10). Assume that the map $\mathbf{w} \mapsto f(\mathbf{w}; \mathbf{z})$ is L -smooth for any $\mathbf{z} \in \mathcal{Z}$ and $L \lesssim n\beta_{S_1, \tilde{A}(S)}$. If

$$T \geq \frac{L_S}{\tilde{\beta}_S} \log \left(nL\beta_{S_1, \tilde{A}(S)} (F_S(\mathbf{w}_0) - \hat{F}_S) / \mu_S \right), \quad (\text{IV.18})$$

then

$$\begin{aligned} & \mathbb{E}[F(A(S)) - \hat{F}_S] \\ & \lesssim \frac{1}{n} \mathbb{E} \left[\frac{\|\nabla f(A(S); \mathbf{z}_1)\|_2^2 + \|\nabla f(A(S); \mathbf{z}'_1)\|_2^2 + 1}{\beta_{S_1, \tilde{A}(S)}} \right]. \end{aligned} \quad (\text{IV.19})$$

Remark IV.13: Note that the assumption $L \lesssim n\beta_{S_1, \tilde{A}(S)}$ is neither restrictive (since n can be very large) nor essential: even without it, a similar result can be obtained by invoking (IV.13) instead of (IV.14). Moreover, although we focus on GD in this section, our analysis applies to any symmetric optimization algorithm. One can directly substitute the corresponding optimization error bounds into Theorem IV.3 or Theorem IV.8 to derive the associated population risk bounds. We omit the details of these extensions for brevity.

V. APPLICATIONS

We now apply the general results obtained for GD to linear models and SNNs. For simplicity, we assume $L \lesssim n\beta_{S_1, \tilde{A}(S)}$. We will demonstrate empirically in the supplementary material that this is a mild assumption.

A. Linear Models

We first consider linear models $\mathbf{x} \mapsto \mathbf{w}^\top \mathbf{x}$, for which the loss function takes the form $f(\mathbf{w}; \mathbf{z}) := \ell_y(\mathbf{w}^\top \mathbf{x})$ for $\mathbf{z} := (\mathbf{x}, y)$. Assume that ℓ_y is μ_ℓ -strongly convex and L_ℓ -smooth for any $y \in \mathbb{R}$, then it can be shown that [17], [18]

$$\beta_{S, \mathbf{w}} \geq \beta'_{S, \mathbf{w}} \geq \mu_\ell \sigma_{\min}(\Sigma_S) \quad \forall \mathbf{w} \in \mathcal{W}, \quad (\text{V.1})$$

where $\sigma_{\min}(\cdot)$ denotes the minimal singular value of a matrix and $\Sigma_S = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i^\top \in \mathbb{R}^{d \times d}$. Therefore, we can choose $\beta_{S, \mathbf{w}} = \mu_\ell \sigma_{\min}(\Sigma_S)$, which satisfies (IV.16) with $L_\beta = 0$. We can also choose $\tilde{\beta}_S = \mu_\ell \sigma_{\min}(\Sigma_S)/2$. Assume $\|\mathbf{x}\|_2 \leq 1$ for any $\mathbf{x} \in \mathcal{X}$, then it is clear that $f(\mathbf{w}; \mathbf{z}) = \ell_y(\mathbf{w}^\top \mathbf{x})$ is L_ℓ -smooth. Recall that L_S is the smoothness parameter of F_S . The following result follows directly from Theorem IV.12 and (V.1). We omit its proof for brevity.

Theorem V.1: Consider the loss function $f(\mathbf{w}; \mathbf{z}) = \ell_y(\mathbf{w}^\top \mathbf{x})$. Let $\tilde{\beta}_S = \mu_\ell \sigma_{\min}(\Sigma_S)/2$ and $L = L_\ell$. Let $\{\mathbf{w}_t\}_{t \in \mathbb{N}}$ be produced by (GD) with $\eta = 1/L_S$ and $A(S) = \mathbf{w}_T$. If T satisfies (IV.18), then we have

$$\begin{aligned} & \mathbb{E}[F(A(S)) - \hat{F}_S] \\ & \lesssim \frac{1}{n} \mathbb{E} \left[\frac{\|\nabla f(A(S); \mathbf{z}_1)\|_2^2 + \|\nabla f(A(S); \mathbf{z}'_1)\|_2^2 + 1}{\mu_\ell \sigma_{\min}(\Sigma_{S_1})} \right]. \end{aligned}$$

Remark V.2: Similar generalization bounds were derived for the empirical risk minimizer (ERM) of linear models [43]. Indeed, if ℓ_y is ρ -Lipschitz and μ_ℓ -strongly convex, then for A being the ERM, it was shown that [43]

$$\mathbb{E}[F(A(S)) - \hat{F}_S] \leq \mathbb{E} \left[\frac{2\rho^2 \text{tr}(\Sigma_S)}{n\mu_\ell \sigma_{\min}(\Sigma_S)} \right],$$

where $\text{tr}(\cdot)$ denotes the trace operator of a square matrix. Theorem V.1 extends the result to GD for linear models, while removing the Lipschitzness and ERM assumptions.

B. Shallow Neural Networks

Consider SNNs with a single hidden layer with d inputs, m hidden neurons, and a single output neuron. Let $\phi : \mathbb{R} \mapsto \mathbb{R}$ be an activation. The prediction function is of the form

$$h_{\mathbf{v}, \mathbf{w}}(\mathbf{x}) = \sum_{k=1}^m v_k \phi(\langle \mathbf{w}^{(k)}, \mathbf{x} \rangle) := \mathbf{v}^\top \Phi(\mathbf{w}\mathbf{x}), \quad (\text{V.2})$$

where $\Phi : \mathbb{R}^m \rightarrow \mathbb{R}^m$ applies ϕ elementwise, $\mathbf{w} := (\mathbf{w}^{(1)}, \dots, \mathbf{w}^{(m)})^\top \in \mathbb{R}^{m \times d}$ and $\mathbf{v} := (v_1, \dots, v_m)^\top \in \mathbb{R}^m$ are the weights of the hidden and output neurons, respectively. Analogous to the work [44], [45], we fix $\mathbf{v}$ as a constant vector with $|v_k| = 1/\sqrt{m}$ and train $\mathbf{w}$ with the mean squared error (MSE) loss $f(\mathbf{w}; \mathbf{z}) = \frac{1}{2}(\mathbf{v}^\top \Phi(\mathbf{w}\mathbf{x}) - y)^2$ on S . We always assume that the feature vectors $\{\mathbf{x}_i\}_{i=1}^n$ have unit ℓ_2 -norm and collect the input data into a data matrix

$$X = (\mathbf{x}_1, \dots, \mathbf{x}_n)^\top \in \mathbb{R}^{n \times d}.$$

For brevity, we always assume $\hat{F}_S = 0$ in this subsection, i.e., there is an SNN with perfect fitting to training examples.

To find a Lipschitz $\tilde{\beta}_{S, \mathbf{w}}$, we impose Assumption V.3 on ϕ , which is widely used in the theoretical analysis of neural networks [45]. Activation functions satisfying Assumption V.3 include the sigmoid and hyperbolic tangent activations.

Assumption V.3: There exist constants $B_1, B_2 > 0$ such that ϕ is B_1 -Lipschitz continuous and B_2 -smooth.

As shown in [46], SNNs defined previously satisfy

$$F_S(\mathbf{w}) \leq \|\nabla F_S(\mathbf{w})\|_F^2 / (2\tilde{\beta}_{S, \mathbf{w}}).$$

Here, $\|\cdot\|_F$ denotes the Frobenius norm of a matrix and

$$\tilde{\beta}_{S, \mathbf{w}} :=$$

$$\frac{1}{n} \sigma_{\min} \left(\frac{1}{m} \sum_{k=1}^m \left(\Phi'(X\mathbf{w}^{(k)}) \left(\Phi'(X\mathbf{w}^{(k)}) \right)^\top \right) \odot (XX^\top) \right), \quad (\text{V.3})$$

where Φ' applies ϕ' elementwise and $\odot$ is the Hadamard (elementwise) product of two matrices. Hence it follows that $\beta_{S, \mathbf{w}} \geq \beta'_{S, \mathbf{w}} \geq \tilde{\beta}_{S, \mathbf{w}}$. Furthermore, we know f is L -smooth with L independent of n and m [14]. The following lemma shows that $\beta_{S, \mathbf{w}}$ is Lipschitz continuous w.r.t. $\mathbf{w}$.

Lemma V.4: Let $\tilde{\beta}_{S, \mathbf{w}}$ be defined in (V.3). If Assumption V.3 holds, then we have

$$|\tilde{\beta}_{S, \mathbf{w}} - \tilde{\beta}_{S, \tilde{\mathbf{w}}}| \leq \frac{2B_1 B_2 \|X\|^2}{n\sqrt{m}} \|\mathbf{w} - \tilde{\mathbf{w}}\|_F.$$

The following theorem gives high probability population risk bounds for SNNs trained by GD. The condition on the number of hidden neurons m is imposed to ensure that the iterates of GD fall inside a local ball around the initial point $\mathbf{w}_0 \in \mathbb{R}^{m \times d}$, which is required to apply a result in [42] that gives linear convergence of the optimization errors for GD. Note this condition is not needed for the stability analysis.

Theorem V.5: Let Assumption V.3 hold. Let $\{\mathbf{w}_t\}_{t \in \mathbb{N}}$ be produced by (GD) with $\eta = 1/L_S$ and $A(S) = \mathbf{w}_T$. Suppose $m \geq 32B_1^2 B_2^2 \|X\|^4 F_S(\mathbf{w}_0) / (n^2 \tilde{\beta}_S^3)$ and T satisfies (IV.18). Define

$$\tilde{\beta}_S := \frac{1}{4n} \sigma_{\min} \left(\mathbb{E}_{\mathbf{w}^{(1)}} \left[\Phi'(X\mathbf{w}^{(1)}) \left(\Phi'(X\mathbf{w}^{(1)}) \right)^\top \right] \odot (XX^\top) \right),$$

where we assume $\mathbf{w}_0^{(k)} \stackrel{\text{i.i.d.}}{\sim} \mathbf{w}^{(1)}$ for each $k \in [m]$. Then with probability (w.r.t. the initialization of $\mathbf{w}_0$) at least $1 - n \exp(-(2B_1^2 \|X\|^2)^{-1} mn \tilde{\beta}_S)$, we have

$$\mathbb{E}[F(A(S))] \lesssim \mathbb{E} \left[\frac{\|\nabla f(A(S); \mathbf{z}_1)\|_2^2 + \|\nabla f(A(S); \mathbf{z}'_1)\|_2^2 + 1}{n \beta_{S_1, \tilde{A}(S)}} \right].$$

Remark V.6: If $n \gtrsim d$, then we expect that $\|X\|$ grows at a rate of $O(\sqrt{n/d})$ [45]. Then the overparameterization condition in Theorem V.5 becomes $m \gtrsim 1/(d^2 \tilde{\beta}_S^3)$. If $n \lesssim d$, then $\|X\|$ is of order of 1 and the overparameterization condition becomes $m \gtrsim 1/(n^2 \tilde{\beta}_S^3)$.

Remark V.7: Linear convergence of GD has been developed for training SNNs [13], [45], which focus on optimization. Theorem V.5 extends these convergence rates to risk bounds, which reflect the behavior of models on testing examples. Stability and generalization of GD were also studied recently for SNNs [14], [15], [47], [48]. Assuming for some $\alpha \in [0, 1]$ that

$$\min_{\mathbf{w}} F(\mathbf{w}) + \lambda \|\mathbf{w} - \mathbf{w}_0\|_2^2 \lesssim \lambda^\alpha \quad (\text{V.4})$$

and $F(\mathbf{w}^*) = 0$, it has been shown that $\mathbb{E}[F(\mathbf{w}_T)] \lesssim n^{-\alpha}$ if $T \asymp n$ and $m \asymp n^3$ [47]. As a comparison, our analysis does not require the assumption in (V.4) and imposes a milder overparameterization condition on m (see Remark V.6). Furthermore, the analysis in [47] derives a sublinear convergence and requires $T \asymp n$ to get good generalization. As a comparison, our analysis uses the linear convergence of GD, which shows that $T \gtrsim 1/\tilde{\beta}_S$ (up to log factors) is enough to get models with good prediction. The analysis [15], [30], [48] requires the loss function to satisfy a self-bounding weak-convexity assumption, which does not hold for the least square loss considered here. Furthermore, their analysis imposes an assumption that the dataset is separable by a neural tangent kernel model with a positive margin, which is not required here. While the analysis in [17] also implies risk bounds of order $O(1/n)$, they require (IV.7) to hold for a β applicable to all $\mathbf{w}$. As a comparison, the risk bounds in Theorem V.5 depend on a much larger quantity $\beta_{S_1, \tilde{A}(S)}$, and therefore could be significantly better.

VI. EXPERIMENTS

We now conduct empirical studies to validate the on-average stability analysis established in our main result, Theorem IV.3. Specifically, we consider shallow neural networks (SNNs) with a single hidden layer containing m neurons. The network is trained using the mean squared error (MSE) loss

$$f(\mathbf{w}; \mathbf{z}) = \frac{1}{2} (h_{\mathbf{w}}(\mathbf{x}) - y)^2. \quad (\text{VI.1})$$

Here, $h_{\mathbf{w}}(\cdot)$ denotes the output of the SNN defined in (V.2) with a fixed $\mathbf{v} = (v_1, \dots, v_m)^\top \in \mathbb{R}^m$, where each v_k is independently sampled from $\{1/\sqrt{m}, -1/\sqrt{m}\}$ with equal probability. To satisfy Assumption V.3, we choose the activation function ϕ to be the sigmoid function. Moreover, $\mathbf{z} := (\mathbf{x}, y) \in \mathbb{R}^d \times \mathbb{R}$ denotes a data sample, where $\mathbf{x} \in \mathbb{R}^d$ is the feature vector and $y \in \mathbb{R}$ is the target variable. Since Theorem IV.3 applies to all symmetric learning algorithms, we train all models using mini-batch SGD. In particular, mini-batch SGD randomly samples indices from the *uniform distribution* over $[n]$ at each iteration. As a result,

TABLE II

SUMMARY OF DATASETS USED IN THE EXPERIMENTS. HERE, ‘ n ’ MEANS SAMPLE SIZE AND ‘ d ’ MEANS NUMBER OF FEATURES.

Dataset	<i>cadata</i>	<i>cpusmall</i>	<i>ijcnn1</i>	<i>covtype</i>
n	20640	8192	49990	581012
d	8	12	22	54
Type	Regression	Regression	Classification	Classification

the ordering of examples in the dataset S does not affect the distribution of the output $A(S)$, since uniform sampling treats all indices identically.

To conduct the experiments, we use four real-world datasets from the LIBSVM library [49], whose information is summarized in Table II. All datasets are preprocessed so that the feature vectors $\mathbf{x}$ are normalized to have unit ℓ_2 -norm. The target variables are rescaled to $y \in [0, 1]$ for regression datasets and $y \in \{0, 1\}$ for binary classification datasets. For each dataset, we randomly split the samples into a training set S containing 80% of the data and a testing set S_{test} containing the remaining 20%.

Our experiments consist of two main parts: (i) we compare the on-average stability ϵ_A defined in (IV.4) with the generalization error estimated from the training and testing datasets; and (ii) we investigate the relationship between the on-average stability ϵ_A at stationary points and sample size n .

A. Comparison Between Stability and Generalization Error

We first compare the on-average stability ϵ_A defined in (IV.4) with the generalization error. As indicated by Lemma III.3, ϵ_A serves as an upper bound on the generalization error. To verify this empirically, we train an SNN with $m = 1000$ hidden neurons on the *cpusmall* and *cadata* datasets using mini-batch SGD. For both datasets, we record the iterates generated with a constant learning rate $\eta = 0.1$ and a batch size of 128. To ensure robustness and statistical reliability, we conduct 40 independent repetitions for each test, with the initial point $\mathbf{w}_0$ in each repetition drawn from a standard Gaussian distribution. To approximately compute ϵ_A in (IV.4), we randomly select one sample from the test set S_{test} as $\mathbf{z}'_1$ and construct a neighboring dataset S_1 by replacing the first sample in S with $\mathbf{z}'_1$. The generalization error is approximated by $F_{S_{\text{test}}}(\mathbf{w}_t) - F_S(\mathbf{w}_t)$, which is an unbiased estimator of the true generalization error.

In Fig. 1, we report the evolution of the mean and standard deviation of the stability ϵ_A and the generalization error as functions of the number of passes (defined as $t \times \text{batchsize}/n$) during training for both datasets. The results show that the stability ϵ_A given in (IV.4) consistently upper bounds the generalization error throughout the training process. Moreover, ϵ_A becomes increasingly close to the generalization error as training progresses. These observations validate Theorem IV.3 and indicate that the bound ϵ_A is relatively tight. In addition, we observe that ϵ_A decreases over the course of training. This behavior can be explained by the fact that SGD tends to converge to a stationary point of the empirical loss F_S , for which $\|\nabla F_S(\mathbf{w}_t)\|_2^2$

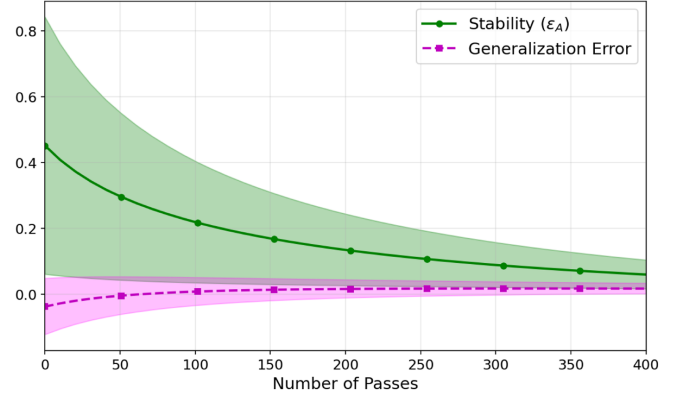
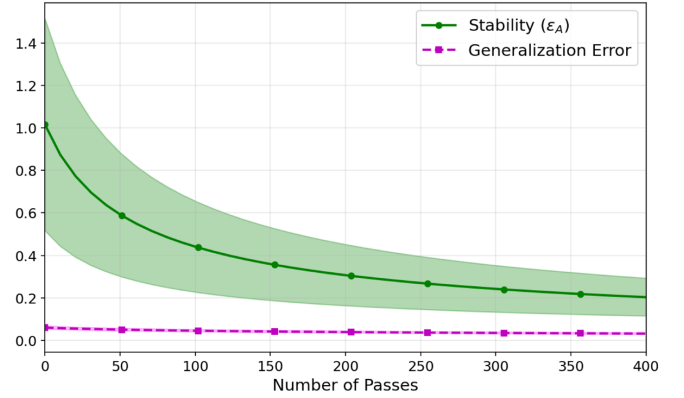
(a) *cpusmall* dataset(b) *cadata* dataset

Fig. 1. Evolution of on-average stability ϵ_A and generalization error during training. Solid and dashed lines record mean values over 40 random repetitions, with shaded regions indicating ± 1 standard deviation.

becomes small as t grows. Consequently, the second term in (IV.4) decreases along the training trajectory.

B. Test on Training Sample Sizes

In this part, we investigate how the stability ϵ_A defined in (IV.4) scales with the number of training samples. As discussed in Remark IV.4, when the optimization algorithm reaches a stationary point, the second term in (IV.4) vanishes, leading to the simplified stability bound in (IV.6), which improves as n increases. We verify this prediction through experiments with varying training set sizes n . Specifically, we use mini-batch SGD to train an SNN model on the datasets *ijcnn1* and *covtype*. For both datasets, the learning rate is set to $\eta = 0.02$ and the batch size is fixed at 32. We consider logarithmically spaced sample sizes n ranging from 1000 to 40000. For each value of n , we run mini-batch SGD until obtaining an iterate $\mathbf{w}_T$ that satisfies the stationarity condition $\|\nabla F_S(\mathbf{w}_T)\|_2 \leq 10^{-6}$. All other experimental settings are the same as in the previous experiment.

Fig. 2 shows how the training sample size n influences the stability ϵ_A at the (nearly) stationary point $\mathbf{w}_T$. We observe that ϵ_A decreases as n increases for both datasets, which is consistent

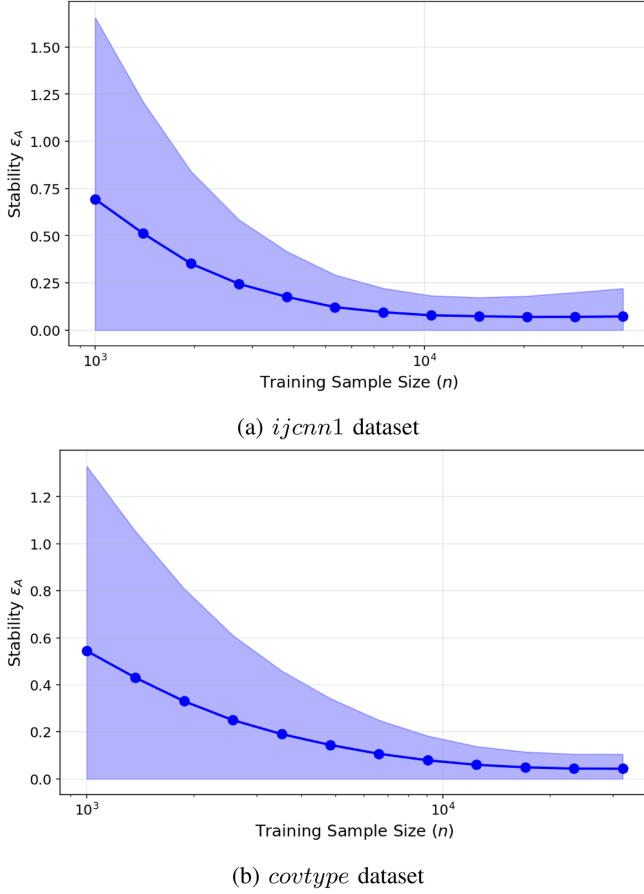


Fig. 2. On-average stability ϵ_A at stationary points as a function of training sample size n . Solid lines show mean values over 40 random repetitions, with shaded regions indicating ± 1 standard deviation.

with our prediction that stability improves with larger training sets.

VII. PROOFS ON MAIN GENERALIZATION RESULTS

In this section, we present the proofs of Theorems IV.3 and IV.8. We start by introducing the following self-bounding property of smooth functions, which shows that the gradient norms can be bounded by the function values.

Lemma VII.1 ([50]): If $g: \mathbb{R}^d \rightarrow \mathbb{R}$ is L -smooth, then

$$\|\nabla g(\mathbf{w})\|_2^2 \leq 2L \left(g(\mathbf{w}) - \min_{\mathbf{w}' \in \mathbb{R}^d} g(\mathbf{w}') \right), \quad \forall \mathbf{w} \in \mathbb{R}^d.$$

Proof of Theorem IV.3: In the proof, we denote by $\mathbb{E}[\cdot]$ the expectation w.r.t. both the datasets S, S' and the learning algorithm A . Given a training set S , it follows from the definitions of F_S and S_i (III.1) that

$$f(A(S_i); \mathbf{z}_i) = nF_S(A(S_i)) - nF_{S_i}(A(S_i)) + f(A(S_i); \mathbf{z}'_i). \quad (\text{VII.1})$$

Since $\mathbf{z}_i$ and $\mathbf{z}'_i$ follow from the same distribution, we have $\mathbb{E}[f(A(S_i); \mathbf{z}'_i)] = \mathbb{E}[f(A(S); \mathbf{z}_i)]$. Hence, taking expectations on both sides of (VII.1), we obtain that

$$\begin{aligned} \mathbb{E}[f(A(S_i); \mathbf{z}_i)] &= n\mathbb{E}[F_S(A(S_i))] \\ &\quad - n\mathbb{E}[F_{S_i}(A(S_i))] + \mathbb{E}[f(A(S); \mathbf{z}_i)], \end{aligned}$$

which together with $\mathbb{E}[F_{S_i}(A(S_i))] = \mathbb{E}[F_S(A(S))]$ due to the symmetry between $\mathbf{z}_i$ and $\mathbf{z}'_i$ further implies

$$\begin{aligned} \mathbb{E}[f(A(S_i); \mathbf{z}_i) - f(A(S); \mathbf{z}_i)] &= n\mathbb{E}[F_S(A(S_i)) - F_{S_i}(A(S_i))] \\ &= n\mathbb{E}[F_S(A(S_i)) - F_S(A(S))]. \end{aligned} \quad (\text{VII.2})$$

On the other hand, we know from the definition of $\beta_{S, \mathbf{w}}$ in (IV.2) that

$$\begin{aligned} \frac{1}{\beta_{S, A(S_i)}} &= \frac{2 \max\{F_S(A(S_i)) - F_S(A(S)), 0\}}{\|\nabla F_S(A(S_i))\|_2^2} \\ &\geq \frac{2(F_S(A(S_i)) - F_S(A(S)))}{\|\nabla F_S(A(S_i))\|_2^2}. \end{aligned} \quad (\text{VII.3})$$

Rearranging the terms in the above inequality gives

$$F_S(A(S_i)) - F_S(A(S)) \leq \frac{\|\nabla F_S(A(S_i))\|_2^2}{2\beta_{S, A(S_i)}},$$

applying which in (VII.2) further implies that

$$\begin{aligned} \mathbb{E}[f(A(S_i); \mathbf{z}_i) - f(A(S); \mathbf{z}_i)] &\leq \frac{n}{2} \mathbb{E}[\|\nabla F_S(A(S_i))\|_2^2 / \beta_{S, A(S_i)}]. \end{aligned} \quad (\text{VII.4})$$

To estimate the RHS of (VII.4), we apply the basic inequality $(a+b)^2 \leq 2(a^2 + b^2)$ twice to obtain

$$\begin{aligned} \|\nabla F_S(A(S_i))\|_2^2 &= \|\nabla F_S(A(S_i)) - \nabla F_{S_i}(A(S_i)) + \nabla F_{S_i}(A(S_i))\|_2^2 \\ &\leq 2\|\nabla F_S(A(S_i)) - \nabla F_{S_i}(A(S_i))\|_2^2 + 2\|\nabla F_{S_i}(A(S_i))\|_2^2 \\ &= 2\left\|\frac{1}{n}\nabla f(A(S_i); \mathbf{z}_i) - \frac{1}{n}\nabla f(A(S_i); \mathbf{z}'_i)\right\|_2^2 \\ &\quad + 2\|\nabla F_{S_i}(A(S_i))\|_2^2 \\ &\leq \frac{4\|\nabla f(A(S_i); \mathbf{z}'_i)\|_2^2}{n^2} + \frac{4\|\nabla f(A(S_i); \mathbf{z}_i)\|_2^2}{n^2} \\ &\quad + 2\|\nabla F_{S_i}(A(S_i))\|_2^2. \end{aligned} \quad (\text{VII.5})$$

Moreover, we know from symmetry between $\mathbf{z}_i$ and $\mathbf{z}'_i$ that

$$\begin{aligned} \mathbb{E}\left[\frac{\|\nabla f(A(S_i); \mathbf{z}'_i)\|_2^2}{\beta_{S, A(S_i)}}\right] &= \mathbb{E}\left[\frac{\|\nabla f(A(S); \mathbf{z}_i)\|_2^2}{\beta_{S_i, A(S)}}\right], \\ \mathbb{E}\left[\frac{\|\nabla f(A(S_i); \mathbf{z}_i)\|_2^2}{\beta_{S, A(S_i)}}\right] &= \mathbb{E}\left[\frac{\|\nabla f(A(S); \mathbf{z}'_i)\|_2^2}{\beta_{S_i, A(S)}}\right], \\ \mathbb{E}\left[\frac{\|\nabla F_{S_i}(A(S_i))\|_2^2}{\beta_{S, A(S_i)}}\right] &= \mathbb{E}\left[\frac{\|\nabla F_S(A(S))\|_2^2}{\beta_{S_i, A(S)}}\right]. \end{aligned}$$

Since $\beta_{S, A(S_i)} \geq 0$ from its definition, dividing both sides of (VII.5) by $\beta_{S, A(S_i)}$ and taking expectations leads to

$$\begin{aligned} \mathbb{E}\left[\frac{\|\nabla F_S(A(S_i))\|_2^2}{\beta_{S, A(S_i)}}\right] &\leq \frac{4}{n^2} \mathbb{E}\left[\frac{\|\nabla f(A(S); \mathbf{z}_i)\|_2^2}{\beta_{S_i, A(S)}}\right] \\ &\quad + \frac{4}{n^2} \mathbb{E}\left[\frac{\|\nabla f(A(S); \mathbf{z}'_i)\|_2^2}{\beta_{S_i, A(S)}}\right] + 2\mathbb{E}\left[\frac{\|\nabla F_S(A(S))\|_2^2}{\beta_{S_i, A(S)}}\right]. \end{aligned}$$

Then it follows from (VII.4) that

$$\begin{aligned} \mathbb{E}[f(A(S_i); \mathbf{z}_i) - f(A(S); \mathbf{z}_i)] &\leq \frac{2}{n} \left(\mathbb{E}\left[\frac{\|\nabla f(A(S); \mathbf{z}_i)\|_2^2}{\beta_{S_i, A(S)}}\right] \right. \\ &\quad \left. + \mathbb{E}\left[\frac{\|\nabla f(A(S); \mathbf{z}'_i)\|_2^2}{\beta_{S_i, A(S)}}\right] \right) + n\mathbb{E}\left[\frac{\|\nabla F_S(A(S))\|_2^2}{\beta_{S_i, A(S)}}\right]. \end{aligned}$$

Taking a summation of the above inequality for $i = 1, \dots, n$, we further obtain

$$\begin{aligned} & \sum_{i=1}^n \mathbb{E} [f(A(S_i); \mathbf{z}_i) - f(A(S); \mathbf{z}_i)] \\ & \leq n \sum_{i=1}^n \mathbb{E} \left[\frac{\|\nabla F_S(A(S))\|_2^2}{\beta_{S_i, A(S)}} \right] + \frac{2}{n} \sum_{i=1}^n \left(\mathbb{E} \left[\frac{\|\nabla f(A(S); \mathbf{z}_i)\|_2^2}{\beta_{S_i, A(S)}} \right] \right. \\ & \quad \left. + \mathbb{E} \left[\frac{\|\nabla f(A(S); \mathbf{z}'_i)\|_2^2}{\beta_{S_i, A(S)}} \right] \right). \end{aligned} \quad (\text{VII.6})$$

Therefore, it follows from Definition III.2 that A is on average ϵ_A -stable with

$$\begin{aligned} \epsilon_A &= \frac{2}{n^2} \sum_{i=1}^n \left(\mathbb{E} \left[\frac{\|\nabla f(A(S); \mathbf{z}_i)\|_2^2}{\beta_{S_i, A(S)}} \right] + \mathbb{E} \left[\frac{\|\nabla f(A(S); \mathbf{z}'_i)\|_2^2}{\beta_{S_i, A(S)}} \right] \right) \\ & \quad + \sum_{i=1}^n \mathbb{E} \left[\frac{\|\nabla F_S(A(S))\|_2^2}{\beta_{S_i, A(S)}} \right] \\ &= \frac{2}{n} \left(\mathbb{E} \left[\frac{\|\nabla f(A(S); \mathbf{z}_1)\|_2^2}{\beta_{S_1, A(S)}} \right] + \mathbb{E} \left[\frac{\|\nabla f(A(S); \mathbf{z}'_1)\|_2^2}{\beta_{S_1, A(S)}} \right] \right) \\ & \quad + n \mathbb{E} \left[\frac{\|\nabla F_S(A(S))\|_2^2}{\beta_{S_1, A(S)}} \right], \end{aligned} \quad (\text{VII.7})$$

where we have used the following identities due to the symmetry of the algorithm A and the fact that the samples follow from the same distribution

$$\begin{aligned} \mathbb{E} \left[\frac{\|\nabla f(A(S); \mathbf{z}_i)\|_2^2}{\beta_{S_i, A(S)}} \right] &= \mathbb{E} \left[\frac{\|\nabla f(A(S); \mathbf{z}_1)\|_2^2}{\beta_{S_1, A(S)}} \right], \forall i \in [n], \\ \mathbb{E} \left[\frac{\|\nabla f(A(S); \mathbf{z}'_i)\|_2^2}{\beta_{S_i, A(S)}} \right] &= \mathbb{E} \left[\frac{\|\nabla f(A(S); \mathbf{z}'_1)\|_2^2}{\beta_{S_1, A(S)}} \right], \forall i \in [n]. \end{aligned}$$

This proves (IV.4).

We proceed to prove (IV.5) using the L_S -smoothness of F_S . In this case, note from Lemma VII.1 and nonnegativity of $\beta_{S_1, A(S)}$ that

$$\mathbb{E} \left[\frac{\|\nabla F_S(A(S))\|_2^2}{\beta_{S_1, A(S)}} \right] \leq 2 \mathbb{E} \left[\frac{L_S(F_S(A(S)) - \hat{F}_S)}{\beta_{S_1, A(S)}} \right].$$

Plugging the above inequality back into (VII.7) gives (IV.5). Finally, the generalization bounds follow directly from Lemma III.3. The proof is now complete. $\square$

Remark VII.2: The above proof also illustrates how the classical PL condition influences stability, as studied in [16], [17]. In particular, by (VII.2) and Definition III.2, the algorithm A is on-average ϵ_A -stable with $\epsilon_A = \sum_{i=1}^n \mathbb{E}[F_S(A(S_i)) - F_S(A(S))]$. For simplicity, here we consider A to be the ERM method. Under the PL condition, we have $\mathbb{E}[F_S(A(S_i)) - F_S(A(S))] \leq \frac{1}{2\beta_S} \mathbb{E}[\|\nabla F_S(A(S_i))\|_2^2]$. Since $\nabla F_{S_i}(A(S_i)) = 0$ by the definition of ERM, it follows that

$$\begin{aligned} & \mathbb{E}[\|\nabla F_S(A(S_i))\|_2^2] \\ &= \mathbb{E}[\|\nabla F_S(A(S_i)) - \nabla F_{S_i}(A(S_i))\|_2^2] \\ &= \frac{1}{n^2} \mathbb{E}[\|\nabla f(A(S_i); \mathbf{z}_i) - \nabla f(A(S_i); \mathbf{z}'_i)\|_2^2]. \end{aligned}$$

Consequently, we have $\epsilon_A \leq \frac{1}{2n^2\beta_S} \sum_{i=1}^n \mathbb{E}[\|\nabla f(A(S_i); \mathbf{z}_i) - \nabla f(A(S_i); \mathbf{z}'_i)\|_2^2]$, which is of order $O(\frac{1}{n\beta_S})$ and highlights how the PL parameter directly controls the stability.

Next, we provide the proof of Theorem IV.8, where we shall frequently apply the following property for an L -smooth function g and $\mathbf{w}, \mathbf{w}' \in \mathcal{W}$:

$$g(\mathbf{w}) \leq g(\mathbf{w}') + \langle \mathbf{w} - \mathbf{w}', \nabla g(\mathbf{w}') \rangle + \frac{L\|\mathbf{w} - \mathbf{w}'\|_2^2}{2}. \quad (\text{VII.8})$$

Proof of Theorem IV.8: Note that $F_S(\tilde{A}(S)) = \hat{F}_S$ from the definition of $\tilde{A}$ in (IV.11). Hence, we can apply (IV.5) with $A = \tilde{A}$ to get

$$\begin{aligned} \mathbb{E} [F(\tilde{A}(S)) - \hat{F}_S] &= \mathbb{E} [F(\tilde{A}(S)) - F_S(\tilde{A}(S))] \\ &\leq \mathbb{E} \left[\frac{2\|\nabla f(\tilde{A}(S); \mathbf{z}_1)\|_2^2}{n\beta_{S_1, \tilde{A}(S)}} \right] \\ &\quad + \mathbb{E} \left[\frac{2\|\nabla f(\tilde{A}(S); \mathbf{z}'_1)\|_2^2}{n\beta_{S_1, \tilde{A}(S)}} \right]. \end{aligned} \quad (\text{VII.9})$$

Moreover, note from the L -smoothness of $f(\cdot; \mathbf{z})$ and Young's inequality that

$$\begin{aligned} & \|\nabla f(\tilde{A}(S); \mathbf{z}_1)\|_2^2 \\ &= \|\nabla f(A(S); \mathbf{z}_1) + \nabla f(\tilde{A}(S); \mathbf{z}_1) - \nabla f(A(S); \mathbf{z}_1)\|_2^2 \\ &\leq 2\|\nabla f(A(S); \mathbf{z}_1)\|_2^2 + 2\|\nabla f(\tilde{A}(S); \mathbf{z}_1) - \nabla f(A(S); \mathbf{z}_1)\|_2^2 \\ &\leq 2\|\nabla f(A(S); \mathbf{z}_1)\|_2^2 + 2L^2\|\tilde{A}(S) - A(S)\|_2^2 \end{aligned}$$

and similarly

$$\begin{aligned} \|\nabla f(\tilde{A}(S); \mathbf{z}'_1)\|_2^2 &\leq 2\|\nabla f(A(S); \mathbf{z}'_1)\|_2^2 \\ &\quad + 2L^2\|\tilde{A}(S) - A(S)\|_2^2. \end{aligned}$$

Hence, it follows from (VII.9) that

$$\begin{aligned} \mathbb{E} [F(\tilde{A}(S)) - \hat{F}_S] &\leq \mathbb{E} \left[\frac{4\|\nabla f(A(S); \mathbf{z}_1)\|_2^2}{n\beta_{S_1, \tilde{A}(S)}} \right. \\ &\quad + \frac{4\|\nabla f(A(S); \mathbf{z}'_1)\|_2^2}{n\beta_{S_1, \tilde{A}(S)}} \\ &\quad \left. + \frac{8L^2\|A(S) - \tilde{A}(S)\|_2^2}{n\beta_{S_1, \tilde{A}(S)}} \right]. \end{aligned} \quad (\text{VII.10})$$

Since F is L -smooth, we have from (VII.8) and the Cauchy-Schwarz inequality that for any $\gamma > 0$

$$\begin{aligned} & F(A(S)) - F(\tilde{A}(S)) \\ &\leq \langle \nabla F(\tilde{A}(S)), A(S) - \tilde{A}(S) \rangle + \frac{L}{2} \|A(S) - \tilde{A}(S)\|_2^2 \\ &\leq \|\nabla F(\tilde{A}(S))\|_2 \|A(S) - \tilde{A}(S)\|_2 + \frac{L}{2} \|A(S) - \tilde{A}(S)\|_2^2 \\ &\leq \frac{1}{4\gamma} \|\nabla F(\tilde{A}(S))\|_2^2 + \left(\gamma + \frac{L}{2} \right) \|A(S) - \tilde{A}(S)\|_2^2. \end{aligned} \quad (\text{VII.11})$$

Recall $\mathbf{w}^* := \arg \min_{\mathbf{w} \in \mathcal{W}} F(\mathbf{w})$. Applying Lemma VII.1 with $\mathbf{w} = \tilde{A}(S)$ implies that

$$\begin{aligned} & \mathbb{E}[F(A(S)) - F(\tilde{A}(S))] \\ & \leq \frac{L}{2\gamma} \mathbb{E} [F(\tilde{A}(S)) - F(\mathbf{w}^*)] + \left(\gamma + \frac{L}{2}\right) \mathbb{E} [\|A(S) - \tilde{A}(S)\|_2^2] \\ & \leq \frac{L}{2\gamma} \mathbb{E} [F(\tilde{A}(S)) - \hat{F}_S] + \left(\gamma + \frac{L}{2}\right) \mathbb{E} [\|A(S) - \tilde{A}(S)\|_2^2], \end{aligned}$$

where the last inequality holds due to $\mathbb{E}[\hat{F}_S] \leq \mathbb{E}[F_S(\mathbf{w}^*)] = F(\mathbf{w}^*)$. Using the above inequality, we can obtain that

$$\begin{aligned} & \mathbb{E}[F(A(S)) - \hat{F}_S] \\ & = \mathbb{E}[F(A(S)) - F(\tilde{A}(S))] + \mathbb{E}[F(\tilde{A}(S)) - \hat{F}_S] \\ & \leq \left(1 + \frac{L}{2\gamma}\right) \mathbb{E} [F(\tilde{A}(S)) - \hat{F}_S] \\ & \quad + \left(\gamma + \frac{L}{2}\right) \mathbb{E} [\|A(S) - \tilde{A}(S)\|_2^2], \end{aligned}$$

which together with (VII.10) further gives

$$\begin{aligned} & \mathbb{E}[F(A(S)) - \hat{F}_S] \\ & \leq \left(\gamma + \frac{L}{2}\right) \mathbb{E} [\|A(S) - \tilde{A}(S)\|_2^2] \\ & \quad + \left(1 + \frac{L}{2\gamma}\right) \mathbb{E} \left[\frac{4\|\nabla f(A(S); \mathbf{z}_1)\|_2^2 + 4\|\nabla f(A(S); \mathbf{z}'_1)\|_2^2}{n\beta_{S_1, \tilde{A}(S)}} \right. \\ & \quad \left. + \frac{8L^2\|A(S) - \tilde{A}(S)\|_2^2}{n\beta_{S_1, \tilde{A}(S)}} \right]. \end{aligned}$$

Plugging $\gamma = L/2$ in the above inequality leads to

$$\begin{aligned} & \mathbb{E}[F(A(S)) - \hat{F}_S] \leq L\mathbb{E} [\|A(S) - \tilde{A}(S)\|_2^2] \\ & \quad + 8\mathbb{E} \left[\frac{\|\nabla f(A(S); \mathbf{z}_1)\|_2^2 + \|\nabla f(A(S); \mathbf{z}'_1)\|_2^2}{n\beta_{S_1, \tilde{A}(S)}} \right. \\ & \quad \left. + \frac{2L^2\|A(S) - \tilde{A}(S)\|_2^2}{n\beta_{S_1, \tilde{A}(S)}} \right], \end{aligned}$$

which proves (IV.12). Additionally, according to the definition of μ_S in (IV.10), we further get

$$\begin{aligned} & \mathbb{E}[F(A(S)) - \hat{F}_S] \leq 2L\mathbb{E} \left[\frac{F_S(A(S)) - \hat{F}_S}{\mu_S} \right] \\ & \quad + 8\mathbb{E} \left[\frac{\|\nabla f(A(S); \mathbf{z}_1)\|_2^2 + \|\nabla f(A(S); \mathbf{z}'_1)\|_2^2}{n\beta_{S_1, \tilde{A}(S)}} \right. \\ & \quad \left. + \frac{4L^2(F_S(A(S)) - \hat{F}_S)}{n\mu_S\beta_{S_1, \tilde{A}(S)}} \right], \end{aligned}$$

which implies (IV.13). The proof is now complete. $\square$

VIII. CONCLUSION

In this paper, we develop a general stability and generalization framework for optimization algorithms in the nonconvex setting. Our generalization bounds depend on an algorithm-dependent

quantity that can be significantly larger than the PL parameter. Moreover, we do not impose restrictive conditions such as the PL inequality and Lipschitzness of loss functions, and thus our results apply to a broad class of learning problems. Additionally, we introduce a one-point QG parameter and derive stability bounds with improved dependence on the optimization error. Our analysis is inspired by the work [16] and relaxes several of the restrictive assumptions therein, such as the QG condition. The resulting theory is highly general and applies to algorithms that do not converge to a global or local minimizer. Instead, one only needs to combine the optimization error bounds of the algorithm with our general stability bound to obtain population risk bounds. We also demonstrate the applicability of our framework by analyzing gradient descent for both linear models and SNNs.

Several open problems remain. First, our current analysis focuses on gradient descent with known linear convergence rates established in the literature [42]; it would be valuable to extend the theory to other optimization algorithms. Second, we derive stability bounds in expectation; developing high-probability bounds would provide deeper insight into the robustness of these algorithms.

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